

Transition of Language in the Catholic Church:
Analysis of Linguistic Shifts in Papal Encyclicals Before and After Vatican II

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Abstract

This study examines linguistic shifts in papal encyclicals before and after the Second Vatican Council (1962-1965), revealing how language in papal encyclical reflects evolving philosophical perspectives within the Catholic Church. Using computational text analysis methods including word frequency analysis, topic modeling, and cosine similarity measures, we identify significant transitions from divine-centered to more human-centered terminology across different pontificates. Our findings demonstrate that the encyclicals of post-Vatican II popes show increased emphasis on social justice, human dignity, and community-oriented themes. The research provides quantitative evidence of how language in official Church documents mirrors broader philosophical developments in contemporary Catholicism.

Keywords: Textual analysis, topic modeling, cosine similarity, word frequency, papal encyclicals, Catholic Church, Second Vatican Council

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Introduction

The Catholic Church has undergone significant linguistic and rhetorical shifts over time, particularly following the Second Vatican Council (Vatican II) in 1962-1965.

Vatican II represented a substantial shift in the Church's relationship with the modern world, often characterized as a transition from an institutionally focused stance to one more engaged with contemporary society and human experience.

While many qualitative theological analyses have explored the doctrinal and pastoral changes following Vatican II (O'Malley, 2008), few studies have employed computational methods to quantitatively analyze how these philosophical shifts manifested in official Church language. Papal encyclicals, authoritative letters addressed to bishops and the Catholic Church in general, serve as ideal texts for such analysis, as they represent the formal theological and philosophical positions of successive popes.

This study employs computational text analysis methods to identify and measure linguistic shifts in papal encyclicals spanning from Pope Leo XIII (1878-1903) to Pope Francis (2013-present). Through word frequency analysis, topic modeling, and cosine similarity measures, I aim to demonstrate how the language of papal documents evolved from a predominantly divine-centered philosophy toward one increasingly concerned with human experience and social justice, themes representative of a contemporary Christianity.

Research Question and Expectations

This study investigates the following primary research questions: First, how has the frequency of human-centered versus divine-centered terminology changed over time in papal encyclicals, particularly before and after Vatican II? Second, what shifts in thematic relationships between key concepts (e.g., faith-reason, divine-human) can be observed across different pontificates? Third, to what extent did the Second Vatican Council represent a linguistic inflection point in papal discourse?

Based on theological accounts of Vatican II's impact and preliminary analyses of word frequencies, I anticipated finding a measurable increase in human-centered terminology in post-Vatican II encyclicals with variations in linguistic patterns among individual popes for each era. These hypotheses were informed by historical accounts of Vatican II as a moment when the Church sought to engage more directly with modern concerns while maintaining continuity with traditional doctrine Alberigo (2006).

Methodology

As reproducibility is essential to science, all code is open-source and can be found on my Github

Data Collection and Preprocessing

The corpus consists of papal encyclicals from Pope Leo XIII (1878-1903) through Pope Francis (2013-present), spanning approximately 150 years. I collected these documents from the Vatican's official archive website using Python's `requests` and `BeautifulSoup4` libraries for web scraping. For efficient document retrieval, I implemented a `ScrapeEncyclicals()` function that systematically navigated the Vatican archives and extracted document URLs. The final corpus included 182 encyclicals, with 93 pre-Vatican II documents (from the pontificates of Leo XIII, Pius X, Benedict XV, Pius XI, and Pius XII) and 89 post-Vatican II documents (from John XXIII, Paul VI, John Paul I, John Paul II, Benedict XVI, and Francis).

The extraction of clean text from Vatican website documents presented significant challenges. Each papal document followed different HTML structures requiring an iterative approach to parsing. I developed a custom `ExtractEncyclicalContent()` function that handled varied document formats through multiple extraction strategies. A particularly difficult aspect was properly identifying document boundaries—many encyclicals contained concluding remarks, editorial notes, or blessing formulas that needed to be properly handled. For instance, ending of the encyclical concluded in various ways to "Given at Rome at St. Peter's on ..." such as differing on the location being given or wording. For instance, some points say "Given in Rome" or "Given at

Rome" or "Given from Rome" or simply "From the Vatican". These differing endings made it difficult, very annoyingly difficult, to identify the correct separator, and so I include many variations. As such, the extraction process required multiple pattern-matching approaches using regular expressions to identify common document components like introductions, conclusions, and section headers. The functions `CleanText()` and `ExtractEncyclicalContent()` handle the text normalization by removing editorial annotations and standardizing document structure across different papal writing styles.

For preprocessing, I utilized NLTK and `spacy` libraries to implement a standardized pipeline. This included removing special characters (`[_";%()|+&=%%. , !?:#$[]/{}<>^`~]`), numbers, and short words ( $\leq 4$  characters), followed by converting all text to lowercase. The texts were then tokenized, lemmatized using NLTK's `WordNetLemmatizer`, and filtered to remove common English stopwords. This preprocessing approach preserved the theological integrity of the content while standardizing the texts for computational analysis.

The quality of extraction and preprocessing was verified through manual inspection of a random sample of documents to ensure accuracy. Despite these efforts, the extraction process remains imperfect given the heterogeneity of document formats across the Vatican archives, particularly for older encyclicals that follow less standardized layouts.

Analytical Methods

Word Frequency Analysis. I calculated relative frequencies of key terms organized into conceptual categories: divine-centered terminology (words such as "catholic," "church," "divine," "sacred") and human-centered terminology (words such as "human," "world," "social," "development"). I calculated relative frequencies of key terms using my `WordFrequencyAnalysis()` function, which normalized word counts by document length and categorized terms into conceptual domains.

Comparative analyses were conducted across pre-Vatican II versus post-Vatican II

papal encyclicals. The frequencies were normalized by document length to ensure fair comparisons across texts of different sizes.

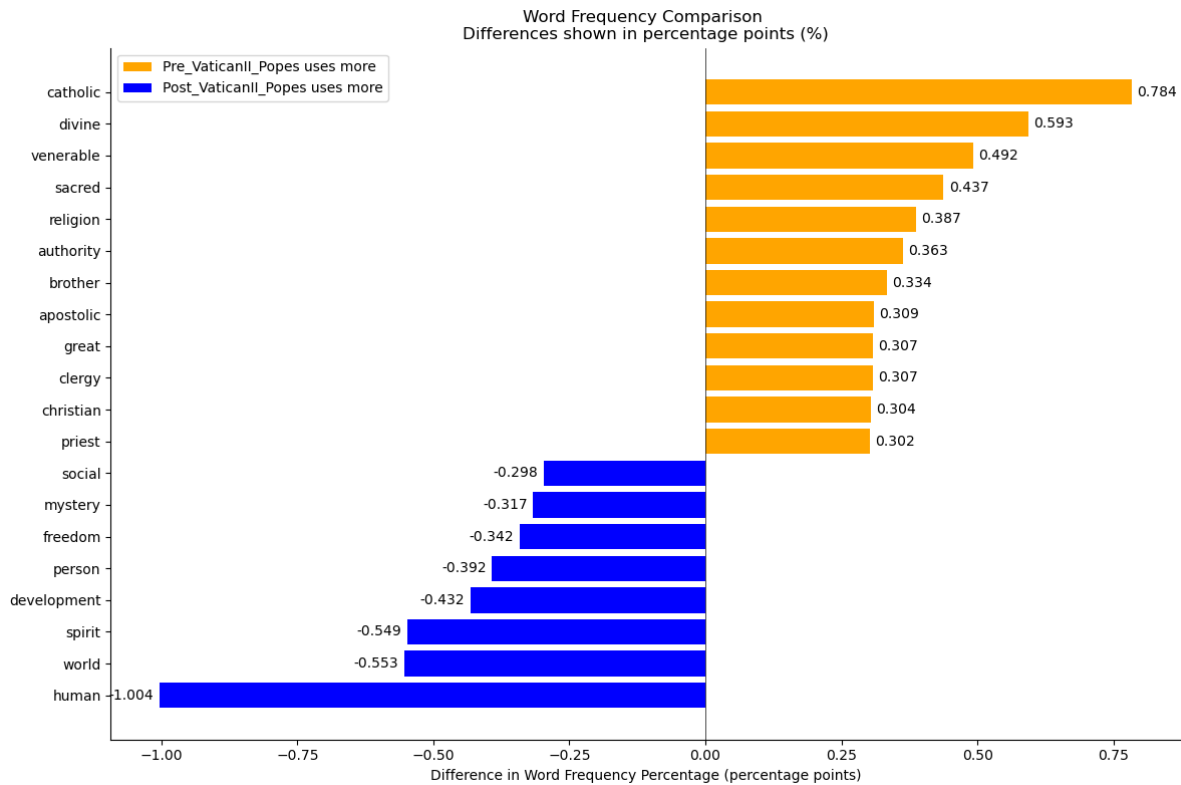


Figure 1. Comparative word frequency analysis of human-centered terminology versus divine-centered terminology can be observed in post-Vatican II papal encyclicals, demonstrating increased human-centered language.

Topic Modeling. I implemented Latent Dirichlet Allocation (LDA) topic modeling using the **Gensim** library to identify broader thematic patterns in the corpus. After experimenting with different parameters, I configured the model to identify 5 topics. This decision followed extensive testing with various configurations; models with fewer topics failed to capture critical theological distinctions, while those with more topics introduced redundancy and diminished interpretability. The five-category framework (Doctrinal, Spiritual, Social, Moral, and Other) offered optimal balance, effectively capturing the core dimensions of Catholic teaching while enabling meaningful cross-pontificate comparison. This configuration proved particularly effective for

quantifying the gradual shift from doctrinal emphasis toward increased social and moral content in post-Vatican II encyclicals, while maintaining sufficient granularity to detect nuanced patterns across the 150-year corpus.

The topic modeling process included creating document-term matrices for each pontificate, fitting LDA models to discover latent topics, calculating topic distribution similarity between pontificates using cosine similarity metrics, and tracking the evolution of topic proportions across different time periods. The function used for this was `TopicModellingFunction()`. Topics were subsequently categorized into broader themes (Doctrinal, Spiritual, Social, Moral, and Other) to track their changing proportions across pontificates.

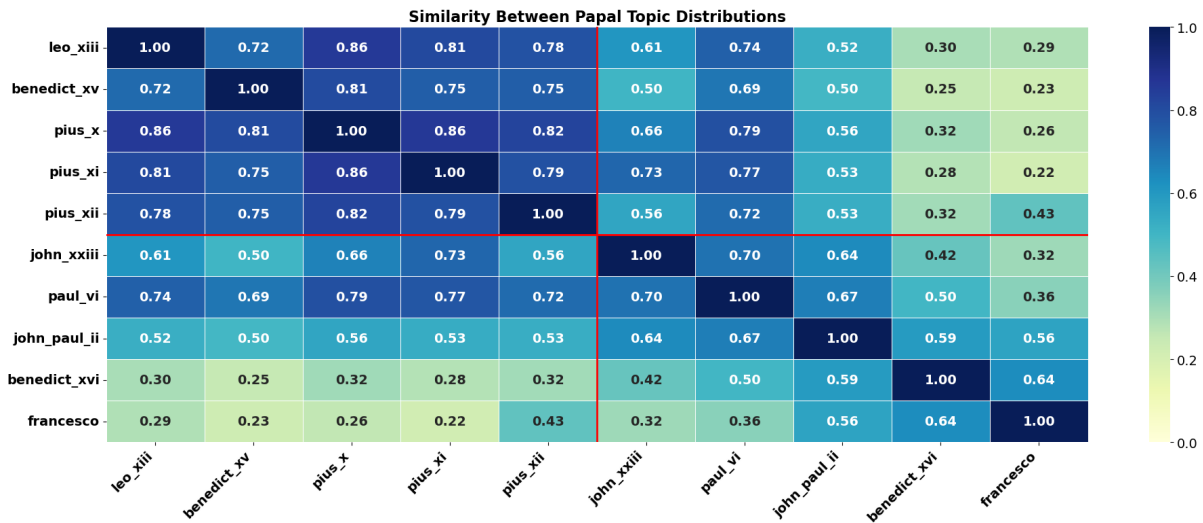


Figure 2. This similarity plot quantifies the cosine similarity between topic distributions across pontificates. It reveals strong thematic cohesion among pre-Vatican II popes (upper left quadrant) and increasing variability in later pontificates (bottom right quadrant), suggesting a gradual rather than abrupt transformation in theological discourse following Vatican II.

Cosine Similarity Analysis. To examine changing relationships between conceptual domains, I employed cosine similarity analysis focused on two key concept pairs: Faith-Reason relationship and Divine-Human relationship. This involved measuring the semantic proximity between terms related to these concept pairs.

The analysis procedure was implemented in Python using NumPy and SciPy

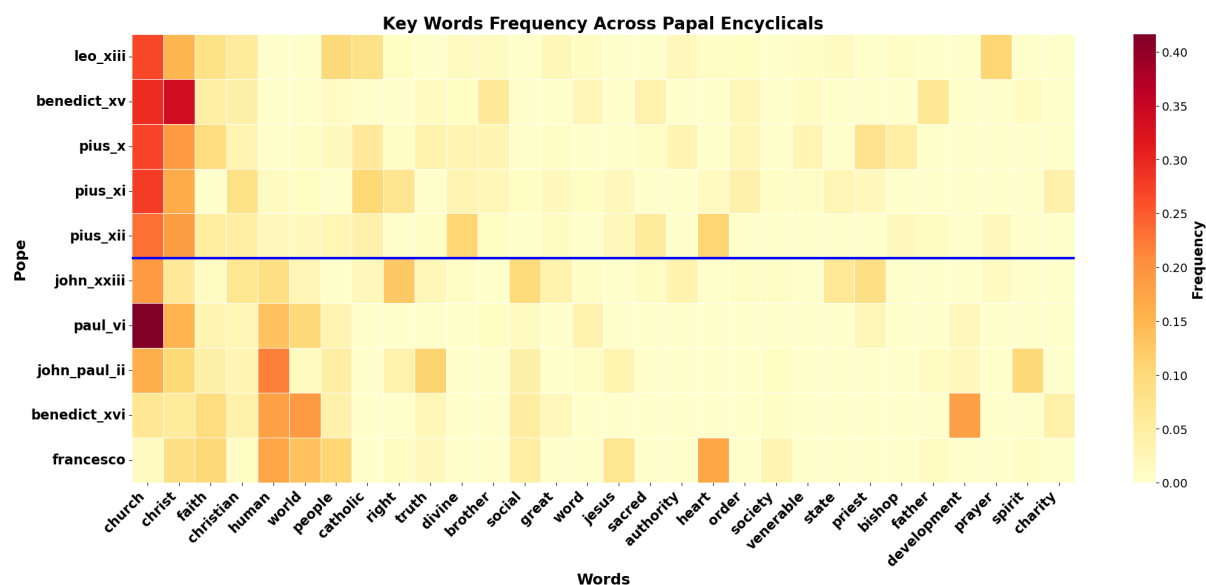


Figure 3. This heatmap visualizes the relative frequency of encyclical keywords across pontificates, revealing distinct vocabulary patterns before and after the Council. Pre-Vatican II encyclicals show greater emphasis on concepts like "church" and "doctrine," while post-Vatican II texts demonstrate increased usage of terms like "human," "social," and "development."

libraries. For each concept pair, I identified word clusters representing each concept, generated context vectors for each concept by examining words appearing within a defined window of concept terms, calculated the cosine similarity between context vectors at different time periods, and applied bootstrapping with 1000 iterations to establish 95% confidence intervals for similarity measures.

This approach allowed me to track not just the frequency of specific terms but the evolving relationship between key theological domains. The window size for context vector creation was set to 100 words after testing different parameters, providing sufficient context while maintaining domain specificity.

These two concepts (Divine-Human and Faith-Reason) are most relative to our narrative. However, the relationship of other concepts may be of the interest to the reader. I have provided these cosine similarity plots in the Appendix. I find that figure 17 on tradition vs. renewal and figure 15 on person vs. community are most interesting due to their significant increase after Vatican II.

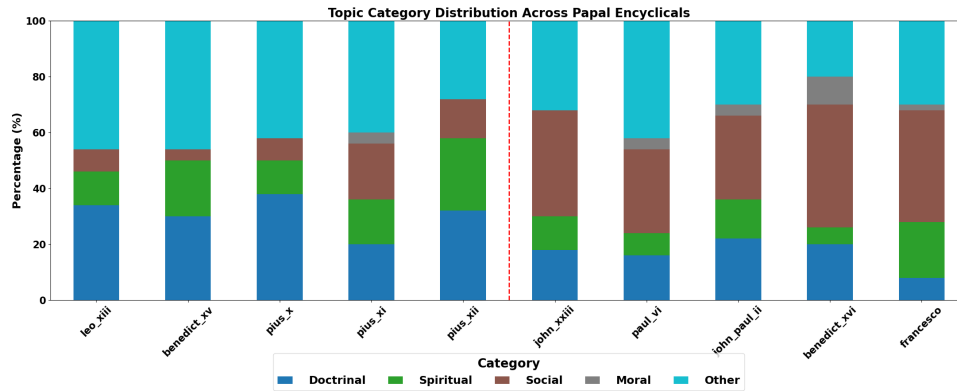


Figure 4. This stacked bar chart displays the proportional distribution of five topic categories (Doctrinal, Spiritual, Social, Moral, and Other) across pontificates calculated with topic model scores. We observe a notable decrease in doctrinal content and corresponding increases in social and moral topics after Vatican II, quantitatively supporting historical interpretations of the Second Vatican Council’s pastoral reorientation.

Results

Word Frequency Analysis Results

The comparative analysis of word frequencies revealed in Figure 1 significant differences in terminology usage between pre- and post-Vatican II encyclicals. Pre-Vatican II encyclicals showed significantly higher frequencies of terms related to institutional authority and divine orientation: "Catholic" (+0.784 percentage points), "Divine" (+0.593 percentage points), "Venerable" (+0.492 percentage points), and "Sacred" (+0.437 percentage points). Post-Vatican II documents exhibited higher frequencies of terms related to human experience and social engagement: "Human" (+1.004 percentage points), "World" (+0.553 percentage points), "Spirit" (+0.549 percentage points), and "Development" (+0.432 percentage points).

For reference, I have also included the word frequency analysis (Figure 8) between the two latest popes: Pope Francis and Benedict XVI, who though are both after Vatican II, often express different views of the church. This word frequency comparison reveals striking stylistic and theological differences between Pope Benedict XVI and

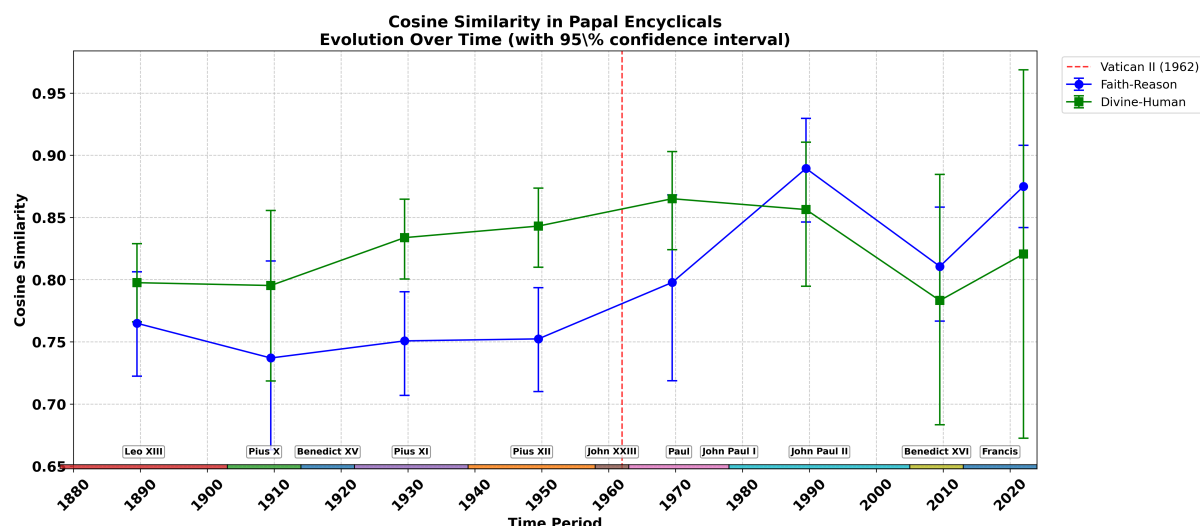


Figure 5. This figure displays the changing cosine similarity between Faith-Reason concepts and Divine-Human concepts. Faith-Reason cosine similarity shows a notable increase after Vatican II with Divine-Human cosine similarity steadily increasing, though before Vatican II. The plot is presented with bootstrapped 95% confidence intervals.

Pope Francis, reflecting their distinct backgrounds and approaches to the papacy.

Francis, the first Latin American pope, demonstrates a marked preference for emotional and relational language (eg. using "heart" significantly more) that emphasizes personal connection. In contrast, Benedict XVI, the German theologian-pope, favors more systematic language (eg development and economic). His vocabulary reflects his European intellectual tradition, focusing on institutional aspects and structured theological reasoning. These linguistic differences highlight how Benedict's papacy centered on doctrinal clarity and theological precision, while Francis emphasizes experiential faith and pastoral accompaniment. This analysis was based on my conversation with John Wilkerson during our February 27th update meeting.

Topic Modeling Results

The LDA topic modeling analysis identified distinct topic distributions across pontificates and revealed changing thematic priorities in papal discourse over time. Table 1 presents the topic models and their labels aggregating pre- and post-Vatican II

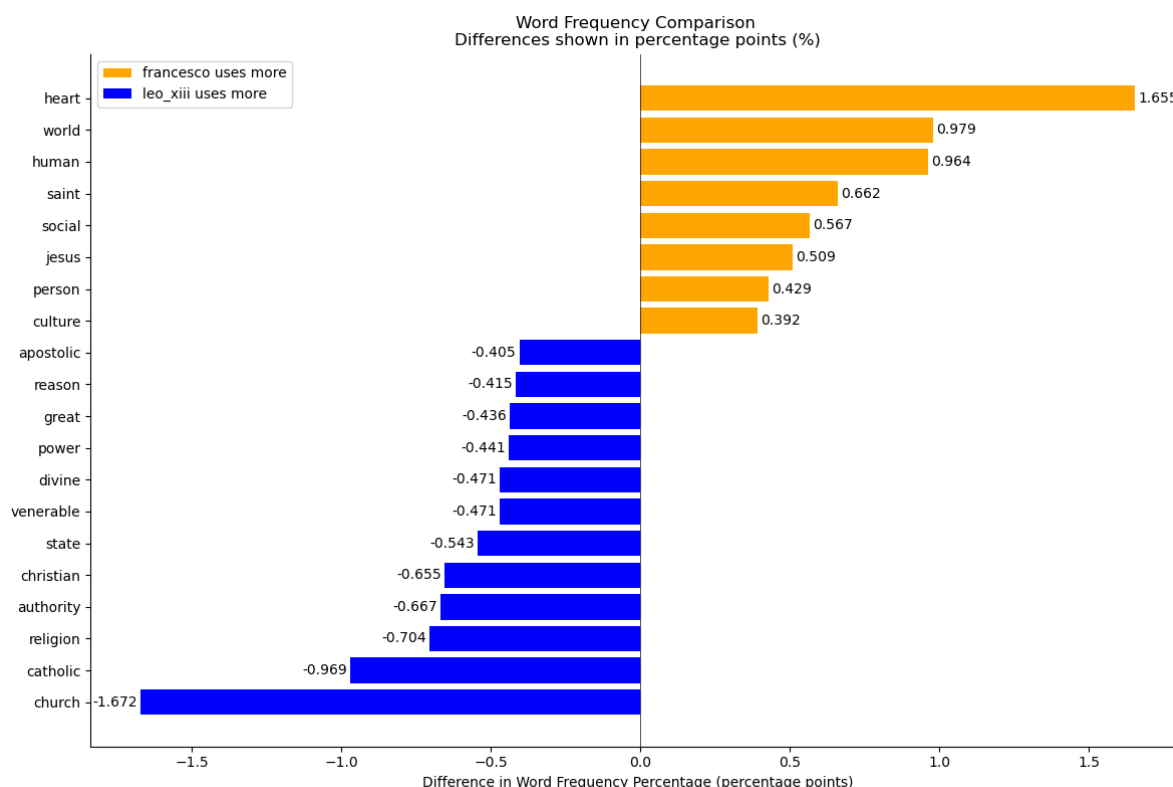


Figure 6. The lowest similarity score from Figure 2 appeared between Pope Leo XIII and Pope Francis. This image shows the difference between the language use to give a deeper insight in those differences. As seen Pope Francis tends to emphasize the human experience (eg. "heart" and "human") while Pope Leo XIII emphasizes the the global church (eg. "church" and "religion.").

popes. The appendix contains topic modelling results for each individual pope.

Strong thematic cohesion emerged among pre-Vatican II popes, with similarity scores ranging from 0.77 to 0.92, while later pontificates showed greater divergence, particularly with Pope Benedict XVI and Pope Francis exhibiting lower similarity to earlier popes. The transition in topic distributions appeared gradual rather than abrupt.

A notable decrease in doctrinal content occurred from pre- to post-Vatican II encyclicals, with corresponding increases in social and moral topics after Vatican II can be seen in Figure ???. This evolution in topic distribution supports historical interpretations of Vatican II's pastoral reorientation, quantitatively demonstrating the Council's impact on papal discourse priorities.

Table 1

This table presents the results of topic modeling analysis comparing pre-Vatican II and post-Vatican II encyclicals. The contrast in key terminology reflects the Council’s influence on Catholic discourse—shifting from institutional/hierarchical themes toward human-centered themes. These linguistic patterns quantitatively demonstrate the Church’s reorientation toward engagement with contemporary society

Era	Top Keywords	Topic Label
Pre-Vatican II	church, catholic, religion, italy, virgin- ity, apostolic, bishop, people, ruthenian, chastity	Authority
	church, spaniard, christ, lycea, college, spain, great, christian, spanish, people	Regional Catholicism
	christ, church, divine, sacred, jesus, heart, christian, mother, prayer, people	Worship
	church, faith, christ, catholic, truth, di- vine, sacred, scripture, authority, great	Orthodoxy
	church, christian, catholic, christ, people, right, state, divine, faith, great	Church-State Relations
Post-Vatican II	human, world, social, development, na- ture, people, environment, nation, prob- lem	Social Development
	spirit, christ, human, church, mercy, fa- ther, world, truth, development, jesus	Spiritual Renewal
	human, right, moral, people, person, world, social, society, freedom, truth	Human Dignity
	church, christ, heart, christian, jesus, faith, spirit, people, council, missionary	Mission
	faith, truth, church, human, christ, world, mother, philosophy, people, reason	Faith and Reason

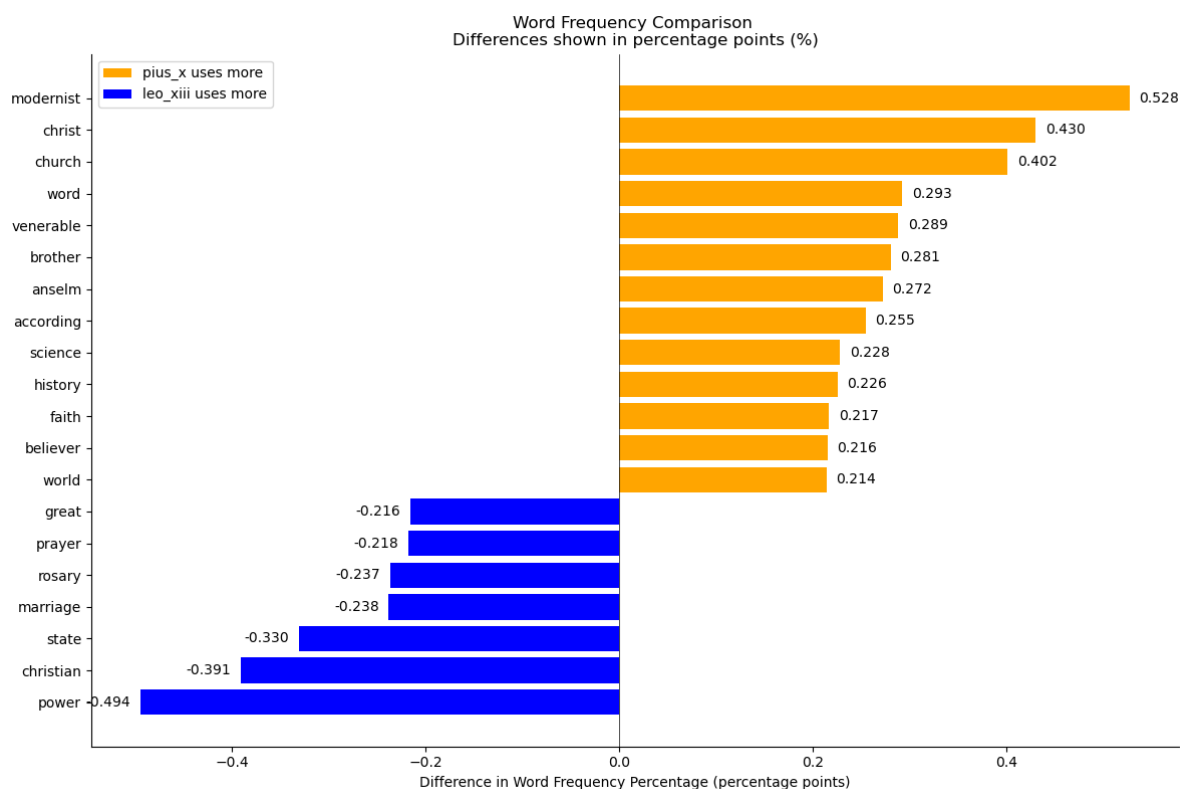


Figure 7. The highest similarity score from Figure 2 appeared between Pope Leo XIII and Pope Pius X. This word frequency comparison demonstrates that even between two pre-Vatican II popes with high similarity scores, differences exist in their rhetorical emphases, with Leo XIII focusing more on church authority and social structures while Pius X directed attention to doctrinal matters and faith elements.

Cosine Similarity Analysis Results

The cosine similarity analysis traced the evolving relationships between faith-reason and divine-human concept pairs over time. For the faith-reason relationship, the analysis showed moderate cosine similarity in the pre-Vatican II period, a notable increase in similarity following Vatican II with values reaching a significant peak (approximately 0.89) during John Paul II's pontificate coinciding with his encyclical "Fides et Ratio [Faith and Reason]" (1998). These findings suggest that Vatican II facilitated a theological reframing that increasingly presented faith and reason as complementary rather than competing domains of knowledge.

For the divine-human relationship, the analysis revealed a gradual increase in

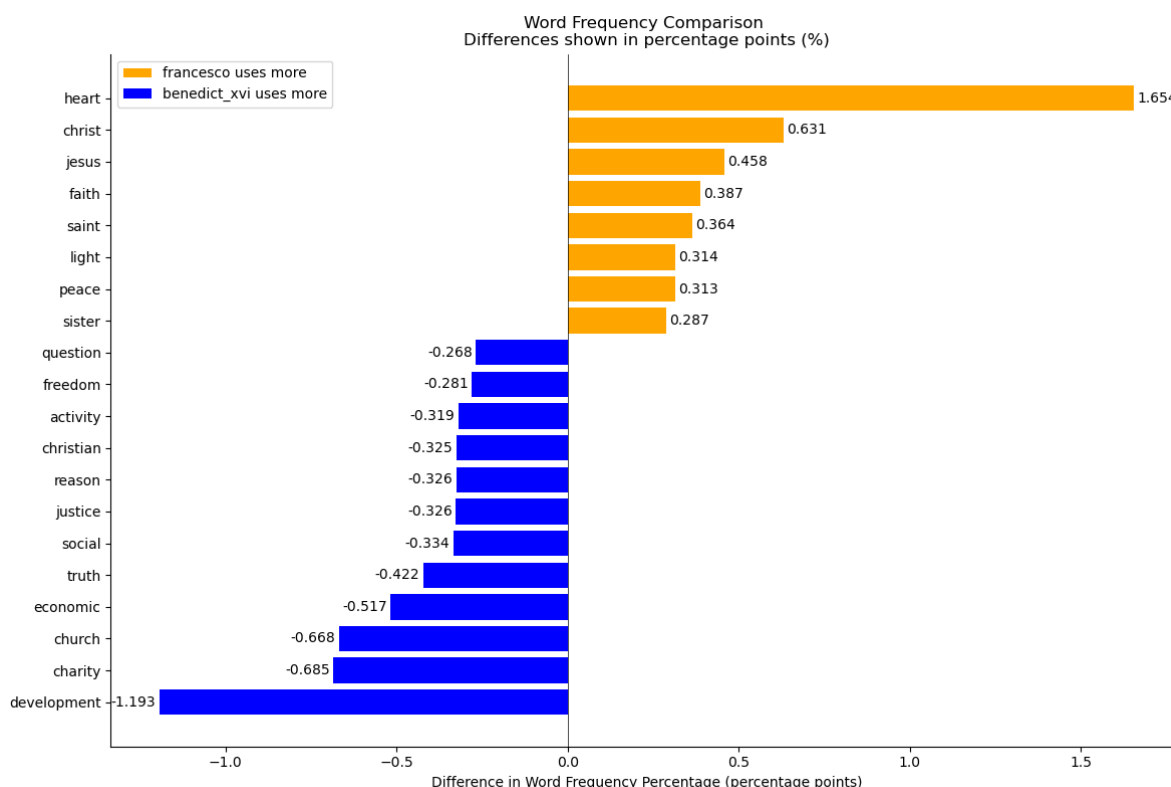


Figure 8. The graph reveals Pope Francis’s significant preference for emotional and relational terminology while Pope Benedict XVI favors developmental and institutional concepts. This reflects their distinct cultural backgrounds and complementary approaches to articulating Catholic teaching despite both being after Vatican II.

similarity beginning before Vatican II, from approximately 0.79 under Leo XIII to 0.84 under Pius XII, and continued increase following Vatican II, reaching peak values (approximately 0.87) during John Paul I’s pontificate. This pattern suggests that the integration of divine and human concerns was already underway before Vatican II, though the Council likely accelerated this conceptual convergence.

To validate the quantitative findings, I identified representative quotes from pre- and post-Vatican II encyclicals that illustrate the shifting conceptualization of these relationships. These can be seen in Table 2. These qualitative examples corroborate the quantitative analysis, demonstrating how theological relationships became increasingly integrated in post-Vatican II discourse.

Discussion

Interpretation of Findings

The combined results provide strong evidence for significant linguistic shifts in papal encyclicals following Vatican II. These shifts reflect broader theological and philosophical developments within the Catholic Church during this period. The word frequency analysis demonstrates a transition from language emphasizing the Church as institution toward language centered on human experience and social engagement. This linguistic shift aligns with Vatican II's pastoral reorientation.

The decreased frequency of terms like "authority," "catholic," and "venerable" coupled with increased usage of "human," "world," and "development" suggests a rhetorical pivot from an ecclesiology emphasizing hierarchical authority toward one more engaged with human concerns. This does not indicate a departure from traditional doctrine but rather a shift in emphasis and presentation.

The cosine similarity analysis reveals a progressive integration of previously more separated conceptual domains, particularly faith-reason and divine-human relationships. This integration suggests a theological evolution toward understanding these domains as complementary rather than competing or hierarchically ordered. The increased semantic proximity between faith and reason concepts following Vatican II aligns with the Council's stated aim to engage more constructively with modern thought.

A key finding across all analytical approaches is that linguistic changes in papal discourse appear evolutionary rather than revolutionary. That is, they are gradual and not sudden. The similarity matrix from topic modeling shows gradual divergence from pre-Vatican II patterns rather than an abrupt break. This pattern supports historical interpretations that emphasize Vatican II as a moment of renewal within a continuous tradition rather than a radical departure (Komonchak, 2009).

Validity of Results

Our computational analysis reveals **significant linguistic shifts** in papal encyclicals, demonstrating a measurable increase in human-centered terminology

following Vatican II while maintaining fundamental doctrinal continuity. The **validity** of these findings is strengthened by the triangulation of three complementary methods (word frequency, topic modeling, and cosine similarity) that converge on similar conclusions, while representative quotes further corroborate these results.

Limitations and Future Directions

This study faces several methodological limitations that suggest directions for future research. The analysis relies on English translations of documents originally written in Latin. While these are official translations, linguistic nuances and theological connotations may not be perfectly preserved across languages. The focus on encyclicals provides a consistent genre for comparison but excludes other important papal documents (apostolic exhortations, apostolic letters, homilies) that might reveal additional dimensions of linguistic change.

Another important limitation of our study is that I only considered the texts of papal encyclicals. Future research may incorporate homily texts to provide a more comprehensive analysis of thematic relationships.

Conclusion

This study has demonstrated significant linguistic shifts in papal encyclicals following Vatican II, providing quantitative evidence for a transition from more divine-centered to more human-centered theological discourse. These shifts manifested in changing word frequencies, topic distributions, and semantic relationships between key conceptual domains.

Methodologically, this project illustrates the value of computational text analysis for studying religious discourse, offering quantitative perspectives that complement traditional theological and historical scholarship. By tracing linguistic patterns across multiple pontificates, I gain insight into how institutional language evolves to reflect changing theological emphases and philosophical frameworks.

This insight suggests that Vatican II represented an important inflection point in Catholic theological discourse, accelerating trends toward greater engagement with

Before and After Vatican II

Faith vs. Reason

Before Vatican II	After Vatican II
<i>Aeterni Patris</i> (1879)	<i>Fides et Ratio</i> (1998)
<i>The great benefits which faith has conferred upon reason: Faith frees and saves reason from error, and endows it with manifold knowledge."</i>	<i>Faith and reason are like two wings on which the human spirit rises to the contemplation of truth; God has placed in the human heart a desire to know the truth."</i>
— Pope Leo XIII	— Pope Saint John Paul II

Divine vs. Humanistic

Before Vatican II	After Vatican II
<i>Immortale Dei</i> (1885)	<i>Deus Caritas Est</i> (2005)
<i>We are bound absolutely to worship God in that way which He has shown to be His will."</i>	<i>In the living community of believers, we experience the love of God. We perceive his presence and we thus learn to recognize that presence in our daily lives."</i>
— Pope Leo XIII	— Pope Benedict XVI

Table 2

These four representative quotes of pre- and post-Vatican II encyclicals corroborate our findings. The linguistic transformation evident in these direct quotations provides validity to the results of our textual analysis.

human experience and social concerns while maintaining continuity with traditional doctrinal foundations. These linguistic changes reflect broader philosophical developments within the Church as it sought to articulate its message in dialogue with modern society.

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Appendix

Table 3

Topic Model Results: Pre-Vatican II Popes (Part 1)

Pope	Top Keywords	Topic Label
Leo XIII	people, church, methodius, cyril, catholic, brother, faith, order, venerable, apostolic	Apostleship
	prayer, rosary, armenian, christ, mother, faith, word, sodality, angel, church	Marian Devotion
	church, christ, faith, catholic, great, people, peter, apostolic, authority, bishop	Church Authority
	christ, church, christian, prayer, faith, divine, mother, rosary, heart, great	Piety
	church, catholic, christian, religion, power, state, authority, right, faith, reason	Church-State Relations
Benedict XV	jerome, christ, church, bible, epist, scripture, truth, christian, word, teaching	Biblical Studies
	church, christ, father, jerome, scripture, christian, faith, sacred, epist, divine	Patristic Heritage
	ephrem, church, father, christ, faith, blessed, sacred, peter, spirit, edessa	Tradition
	christ, church, brother, order, apostle, apostolic, preaching, people, venerable, peace	Apostolic Mission
	christ, church, jerome, brother, scripture, epist, word, christian, faith, sacred	Biblical Interpretation

Table 4

Topic Model Results: Pre-Vatican II Popes (Part 2)

Pope	Top Keywords	Topic Label
Pius X	christ, church, catholic, faith, christian, brother, order, venerable, divine, people	Doctrinal Orthodoxy
	church, christ, truth, christian, order, faith, gregory, people, divine, world	Christian Truth
	church, christ, faith, catholic, divine, truth, venerable, bishop, authority, great	Church Authority
	priest, bishop, woman, obedience, christ, authority, faithful, church, venerable, brother	Hierarchy
	church, faith, christ, modernist, catholic, religion, brother, authority, sacred, right	Anti-Modernism
Pius XI	church, catholic, christ, right, christian, order, state, great, society, heart	Social Order
	church, catholic, order, christ, right, chris- tian, action, social, great, brother	Catholic Action
	church, christ, christian, right, state, catholic, divine, priest, brother, venerable	Church-State
	little, charity, child, jesus, predecessor, es- pecially, soul, word, brother, heart	Charitable Works
	christ, church, christian, divine, right, hu- man, catholic, world, priest, family	Society

Table 5

Topic Model Results: Vatican II Transition

Pope	Top Keywords	Topic Label
Pius XII	nature, queen, christ, council, faith, person, father, called, bishop, human	Marian Theology
	church, sacred, catholic, divine, christian, faith, great, people, bishop, religious	Sacred Tradition
	church, christ, peace, heart, world, divine, human, brother, truth, people	Fellowship
	church, christ, divine, faith, christian, sacred, catholic, prayer, jesus, people	Christian Faith
	heart, divine, christ, virgin, church, christian, mother, jesus, devotion, prayer	Sacred Heart Devotion
John XXIII	social, economic, human, right, world, nation, state, principle, individual, country	Social Teaching
	priest, vianney, great, church, christ, prayer, sacred, soul, word, divine	Vocation
	church, christian, christ, catholic, unity, truth, faith, world, great, apostolic	Ecumenical Unity
	church, human, christian, right, christ, unity, social, truth, state, authority	Human Rights
	right, state, authority, common, human, nation, public, duty, society, order	Common Good

Table 6

Topic Model Results: Post-Vatican II Popes (Part 1)

Pope	Top Keywords	Topic Label
Paul VI	church, christ, eucharist, word, faith, mystery, sacrament, present, sacrifice, council	Worship
	church, christ, nation, world, human, present, people, christian, dialogue, word	Global Dialogue
	church, christ, world, priest, dialogue, celibacy, christian, human, spiritual, faith	Priestly Ministry
	nation, human, world, peace, development, people, social, effort, church, brother	International Development
	church, human, married, family, moral, husband, marriage, natural, encyc, world	Marriage and Family
John Paul II	human, social, right, people, development, world, church, society, economic, worker	Social Justice
	church, christ, christian, missionary, people, council, spirit, faith, community, communion	Church Mission
	human, moral, church, truth, person, freedom, christ, jesus, people, right	Human Dignity
	spirit, christ, church, jesus, mother, mystery, father, human, faith, word	Theology
	truth, philosophy, human, faith, reason, philosophical, knowledge, thought, theology, revelation	Faith and Reason

Table 7

Topic Model Results: Post-Vatican II Popes (Part 2)

Pope	Top Keywords	Topic Label
Benedict XVI	world, faith, christ, human, christian, great, freedom, suffering, present, life	Christianity
	development, human, church, social, charity, world, people, faith, christ, christian	Integral Development
	world, faith, christ, life, human, christian, people, great, freedom, truth	Christian Freedom
	development, human, church, social, charity, economic, people, truth, world, society	Economics
	human, development, world, church, social, charity, faith, truth, people, christian	Charitable Development
Francis	human, world, environment, social, nature, creature, problem, creation, environmental, people	Environmental Concern
	faith, light, christ, jesus, truth, church, saint, christian, history, world	Light of Faith
	people, human, world, society, social, person, right, culture, dignity, individual	Human Dignity
	heart, christ, jesus, saint, human, devotion, father, world, desire, spirit	Christian Heart
	heart, world, human, jesus, faith, christ, people, saint, society, social	Social Compassion

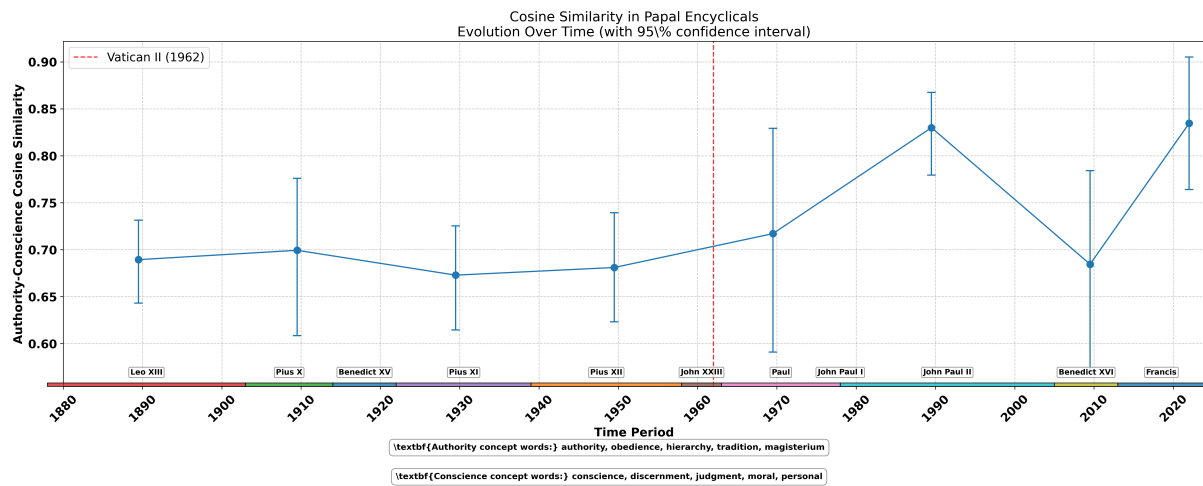


Figure 9. Cosine similarity between authority and conscience related concepts in papal encyclicals (1880-2020), showing an increasing trend after Vatican II.

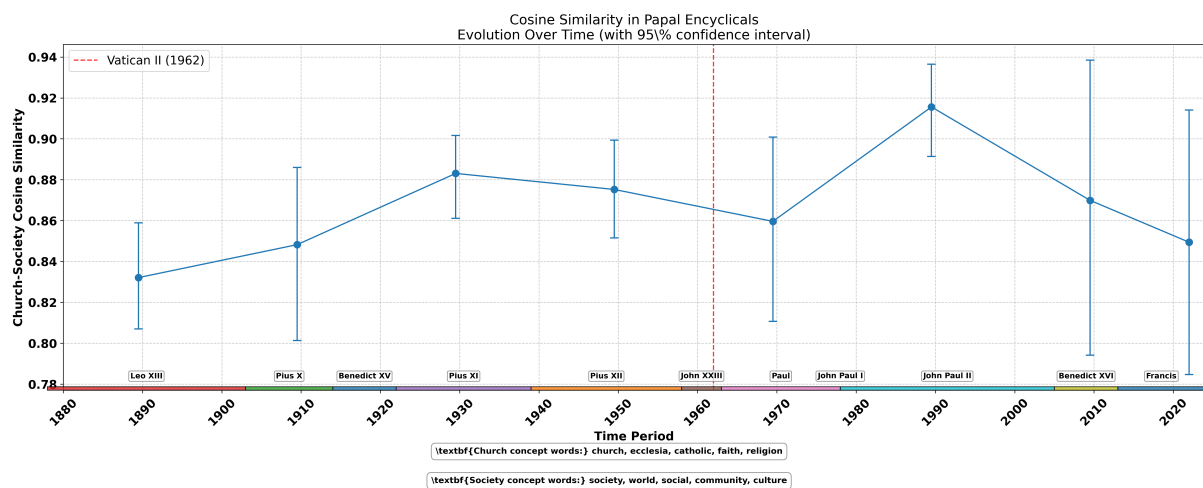


Figure 10. Cosine similarity between church and society related concepts in papal encyclicals (1880-2020), showing a gradually rising similarity score.

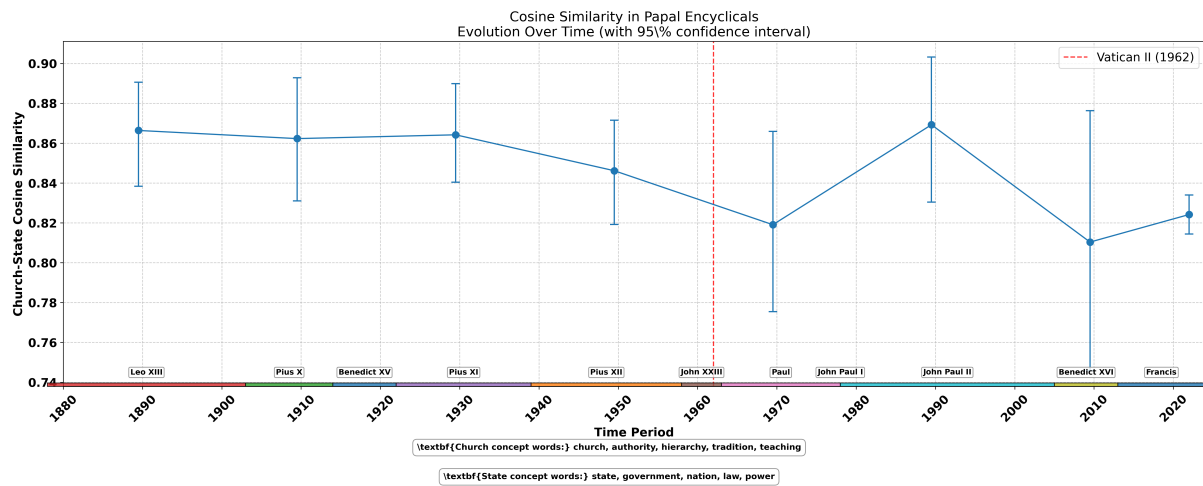


Figure 11. Cosine similarity between church and state related concepts in papal encyclicals (1880-2020) with a gradual and slow decline.

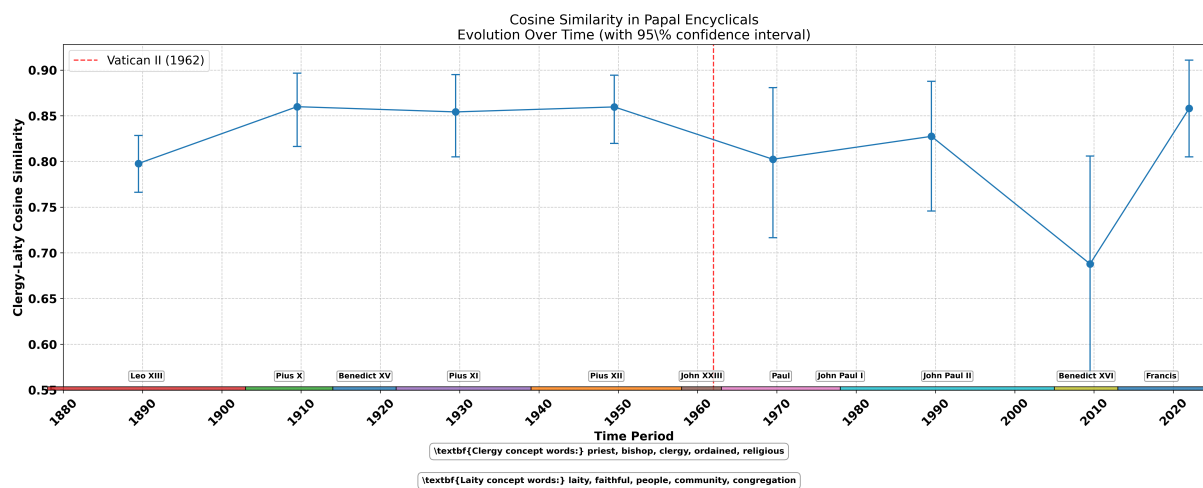


Figure 12. Cosine similarity between clergy and laity related concepts in papal encyclicals (1880-2020) with a gradual and slow decline.

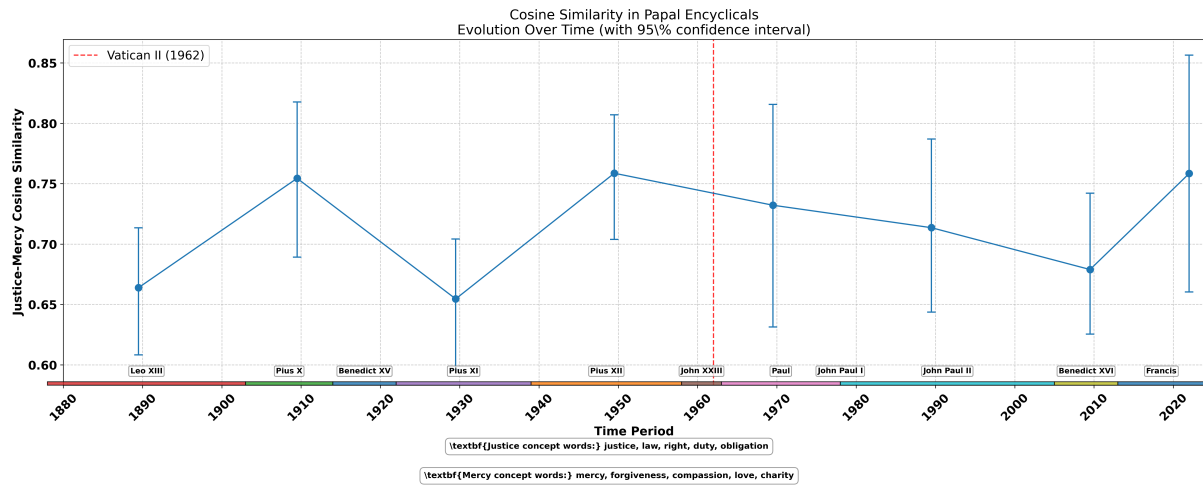


Figure 13. Cosine similarity between justice and mercy related concepts in papal encyclicals (1880-2020).

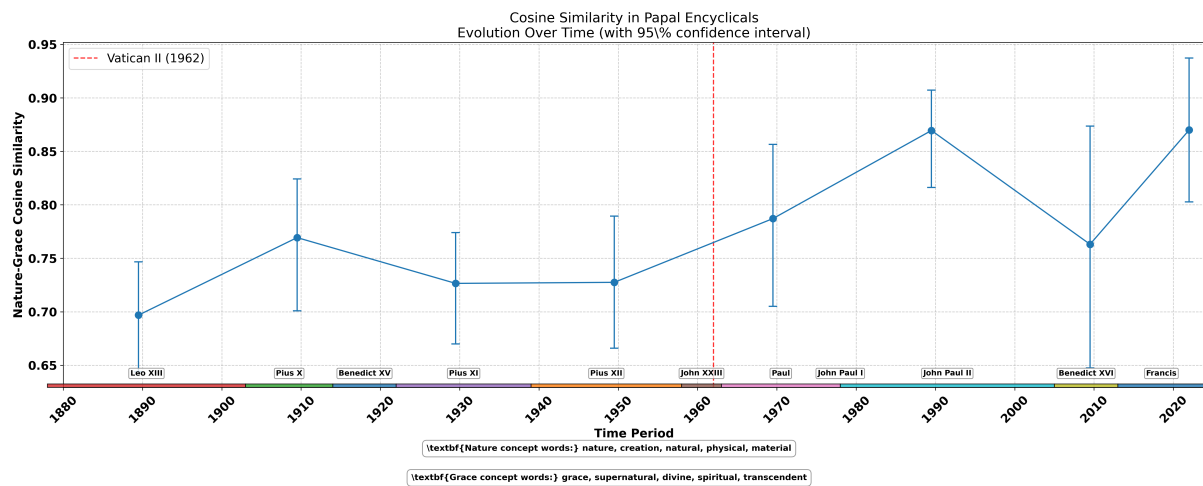


Figure 14. Cosine similarity between nature and grace related concepts in papal encyclicals (1880-2020), illustrating a significant upward trend after Vatican II with notable peaks during John Paul II and Francis's pontificates.

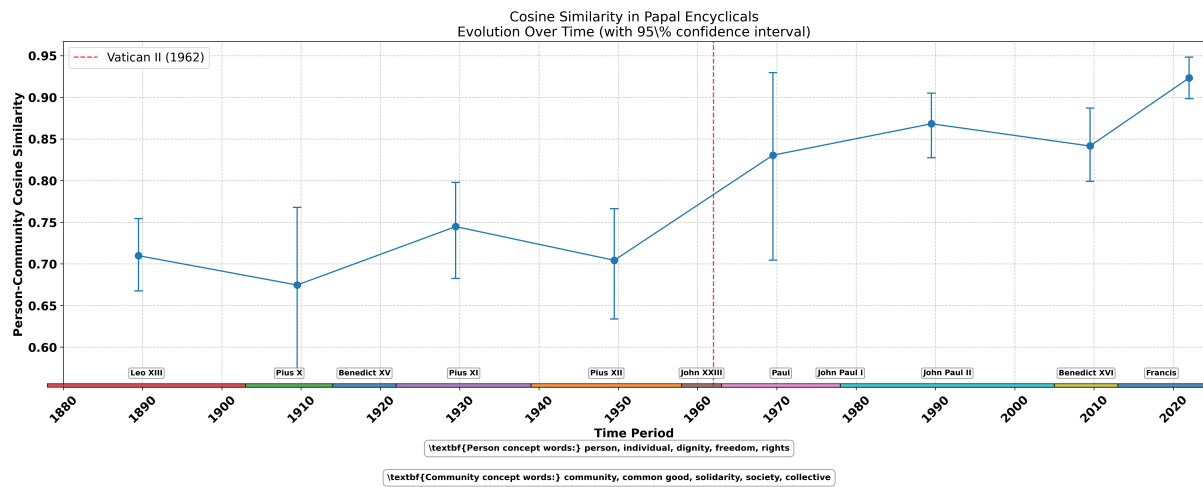


Figure 15. Cosine similarity between person and community related concepts in papal encyclicals (1880-2020), showing a substantial increase after Vatican II with the highest values during John Paul II and Francis's papacies.

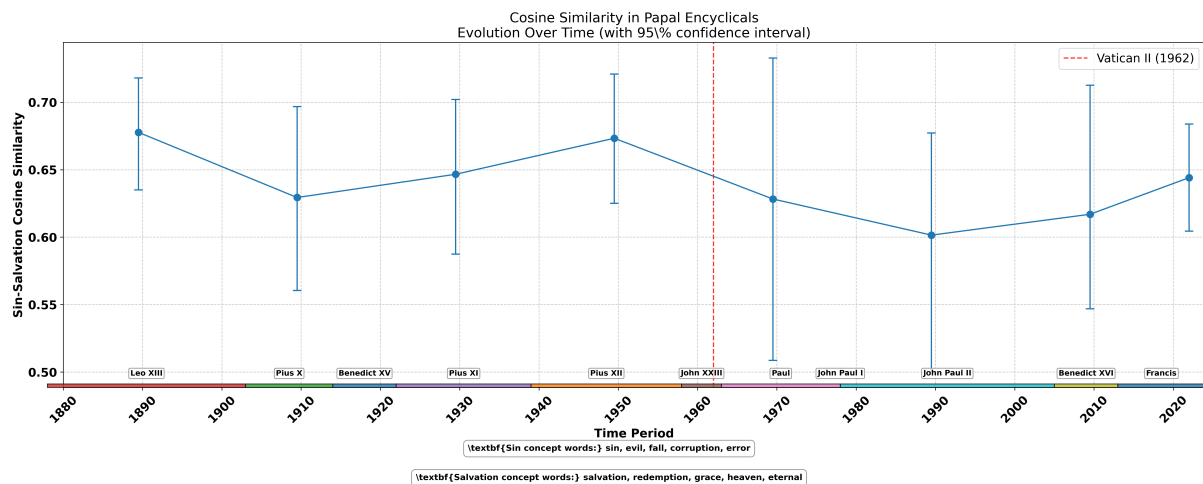


Figure 16. Cosine similarity between sin and salvation related concepts in papal encyclicals (1880-2020), displaying the lowest overall similarity compared to other concept pairs with minimal change over time.

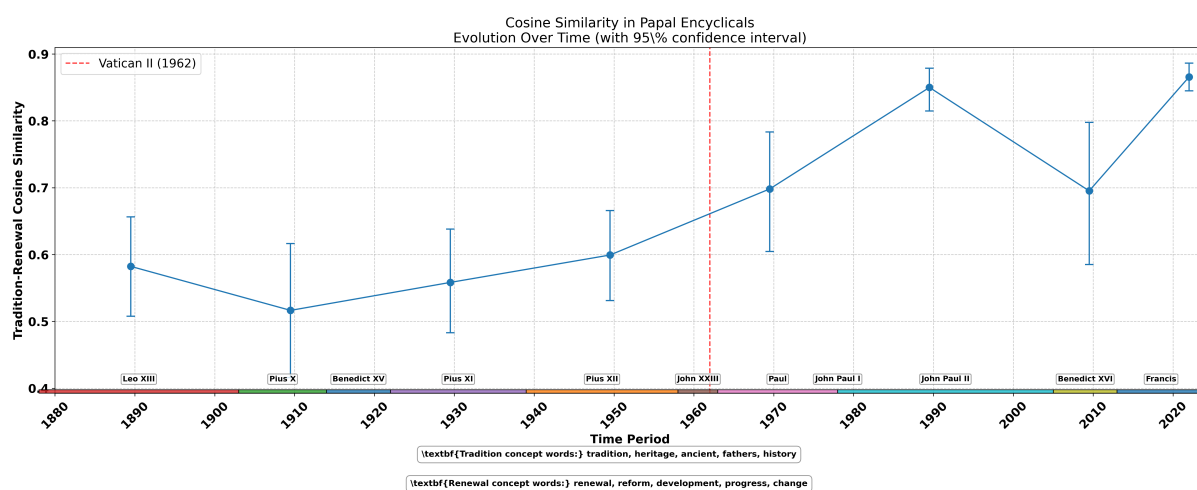


Figure 17. Cosine similarity between tradition and renewal related concepts in papal encyclicals (1880-2020), revealing a dramatic increase after Vatican II with peaks during John Paul II and Francis's pontificates.